16. $K_a = 1 \times 10^{-3}$

$pH = 4$

$\frac{[salt]}{[Acid]} = ?$

$pH = pK_a + \log \frac{[Salt]}{[Acid]}$

$4 = -\log_{10}(1 \times 10^{-3}) + \log \frac{[Salt]}{[Acid]}$

$4 = 3 + \log \frac{[Salt]}{[Acid]}$

$1 = \log_{10} \frac{[Salt]}{[Acid]}$

$\Rightarrow \frac{[Salt]}{[Acid]} = 10^1$

i.e., $\frac{[Acid]}{[Salt]} = \frac{1}{10}$

1:10

[Option (d)]

17. $\underset{10^{-5}m}{KOH} \rightarrow \underset{10^{-5}m}{K^+} + \underset{10^{-5}m}{OH^-}$

$[OH^-] = 10^{-5}M.$

$pH = 14 - pOH$

$pH = 14 - \left(-\log \left[OH^-\right]\right)$

$= 14 + \log [OH^-]$

$= 14 + \log 10^{-5}$

$= 14 - 5$

$= 9.$

[Option (a)]

18. $\underset{\text{acid 1}}{H_3PO_4} + \underset{\text{base 1}}{H - OH} \rightleftharpoons \underset{\text{acid 2}}{H_3O^+} + \underset{\text{base 2}}{H_2PO_4^-}$

$\therefore H_2PO_4^-$ is the conjugate base of H_3PO_4

[Option (c)]

19. HPO_4^{2-} can have the ability to accept a proton to form $H_2PO_4^-$.

It can also have the ability to donate a proton to form PO_4^{-3}

[Option (c)]

20. $pH = -\log_{10}[H^+]$

$\therefore [H^+] = 10^{-pH}$

$= 10^0 = 1$

$[H^+] = 1M$

The solution is strongly acidic

[Option (b)]

21. According to Henderson equation

$pH = pK_a + \log \frac{[salt]}{[acid]}$

ie. $-\log [H^+] = -\log K_a + \log \frac{[salt]}{[acid]}$

$-\log[H^+] = \log \frac{[salt]}{[acid]} \times \frac{1}{K_a}$

$\log \frac{1}{[H^+]} = \log \frac{[salt]}{[acid]} \times \frac{1}{K_a}$

$\therefore [H^+] = K_a \frac{[acid]}{[salt]}$

[option (a)]

22. $h = \sqrt{\frac{K_h}{K_a . K_b}}$

[Option (c)]

23. $K_h = \frac{K_w}{K_b} = \frac{1 \times 10^{-14}}{1.8 \times 10^{-5}}$

$= 0.55 \times 10^{-9}$

$= 5.5 \times 10^{-10}$

[option (b)]

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